

**EFFECT OF NEEM EXTRACT ON LEAFHOPPER(*AMRASCA
BIGUTTULA BIGUTTULA*) OF POTATO CROP(*SOLANUM
TUBEROSUM* L.)AT QUETTA, BALOCHISTAN**



BY

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**DEPARTMENT OF ENTOMOLOGY
FACULTY OF CROP PROTECTION
TO SINDH AGRICULTURE UNIVERSITY TANDOJAM
2016**

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**A THESIS SUBMITTED THROUGH THE DEPARTMENT OF ENTOMOLOGY
FACULTY OF CROP PROTECTION, TO SINDH AGRICULTURE UNIVERSITY
TANDOJAM IN CONNECTION WITH THE PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE DEGREE OF
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RESEARCH CERTIFICATE

Certified that the present research work entitled “**Effect of neem extract on leafhopper(*Amrasca biguttula biguttula*) of potato crop (*Solanum tuberosum* L.) at Quetta, Balochistan**” embodied in this thesis has been carried out by **Nastullah Mastoi**, under my guidance and supervision in connection with the partial fulfillment of the requirements for the degree of Master of Science (Agriculture) Honours in Entomology, and that the work is original.

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THESIS RELEASE FORM

I, **Nasrullah Mastoi** hereby authorize Sindh Agriculture University, Tandojam, to supply copies of my thesis to libraries or individuals upon request.

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BY

M A S T O I

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AN ABSTRACT OF THE THESIS OF

NASRULLAH MASTOI

For MASTER OF SCIENCE (AGRI.)
HONS.

Major ENTOMOLOGY

TITLE: EFFECT OF NEEM EXTRACT ON LEAFHOPPER (*AMRASCA BIGUTTULA BIGUTTULA*) OF POTATO CROP (*SOLANUM TUBEROSUM* L.) AT QUETTA, BALUCHISTAN.

Field studies on efficacy of neem extract against Leafhopper (*Amrasca biguttula biguttula*) on potato crop (*Solanum tuberosum* L.), was carried out at the Insect Systematic Postgraduate Laboratory, Department of Entomology, Sindh Agriculture University Tandojam and Quetta Baluchistan, during the year of 2015. The crop was sown on well prepared seed bed in a Randomized Complete Block Design keeping a net treatment plot area of 5x4 meter with 4 replications and 4 treatments. The pre-treatment observations were taken 24hrs before spray, whereas, post-treatment observations were taken at the intervals of 24hrs, 48hrs and 72hrs after the respective spray. Relative neem extract displayed greater reduction of leafhopper (*A. b. biguttula*).

For the identification of potato leafhopper (*Amrasca biguttula biguttula*) species, they were collected and studied at postgraduate systematic laboratory, S. A. U. Tandojam. Habitus images were captured through microscope, further for the study of genitalia, maceration was undertaken to remove soft bodies and connective tissues to have clear genitalia for study. The study revealed it *A. b. biguttula* (Ishida, 1912) the detailed hierarchy is given below and habitus images and genitalia are provided in Plate-1. The results revealed that all the sprays of neem extract against potato leafhopper (*A. b. biguttula*) remained effective after the observation of leafhopper (*A. b. biguttula*) population on 24hrs, 48hrs and 72hrs after the respective spray. T₄=100cc/plot, showed highest efficacy (59.31a %) through out the three sprays followed by T₃=75cc/plot (50.71b %), T₂=50cc/plot (43.23c %) and T₁=25cc/plot (43.23d %) respectively, with comparison to T₅=Control/untreated (7.09e %).

CHAPTER-I

INTRODUCTION

Potato (*Solanum tuberosum*L.) well known as staple food and vegetable crop. It ranked 4th position among other after cereals crops (Rauscher *et al.*, 2006). The crop relatively higher in its production capacity and per unit calories compared to wheat and rice, it produced 12-15 times more yield ha-1 with 1.9 metric tons GDP compared to other cereal grains crops (GOP, 2011). The crop mostly grown in summer around 03 consecutive crops can be grown per year in plains and low valleys

According to the map, there is 08 agro ecological zone of potato production includes (1) Southern Sindh, Punjab and Balochistan, (2) South East KPK, Central Punjab, (3) KPK & Northern Punjab (4) KPK irrigated areas, (5) KPK rain fed valleys & hill sides, AJK & Punjab Northern (6) KPK irrigated high valleys, Northern AJK and Chillas, (7) Northern KPK irrigated high valleys and Mastuj, (8) Balochistan irrigated high valleys and North Waziristan, respectively. All these zones are variables in their respective climate, soil, irrigation, latitude, longitude and altitudes (Zanoni, 1991).

Nevertheless, the insect pests cause 35-40% losses that may reach up to 60-70% (Salim, 1999). Potato crop (*Solanum tuberosum*L.) is attacked by different insect pests among them the potato leaf hoppers (*A. b.biguttula*) possess considerable importance. About 79 species of leaf hopper have been found infesting the potato crop, *Solanum*

tuberosum L. (Ramirez and Ramos, 1982). The damage mode action relies on climatic factors and host availability. Pesticides are quick solutions for insect control and are important in increasing solutions for insect control and also important in increasing agricultural production (Mehmood *et al.*, 2001). In fruits and vegetable crops there was around 1/3 pesticides applied in the country, which prevent the damaged of crop and dramatically improved the yield (Hussain *et al.*, 2002).

Potato leafhopper (*A. b. biguttula*) 1/8 inch long, light green, sedge-shaped and slender appearance during immature and adult stages. Both adults and nymphs have piercing sucking mouthparts on the underside of the head that they insert into the plant to feed on plant juices. The wingless nymphs move quickly backwards and sideways in a crab like fashion on the leaf surface. Eggs are laid in plant tissue and are difficult to observe in the field. Potato leafhopper, *A. b. biguttula*. (Homoptera: Cicadellidae), is a highly polyphagous pest, capable of successful reproduction on 200 plant species in 25 families (Lamp *et al.* 1994).

In view of the environmental and health issues the bio-pesticides are holding attention by scientists and researchers as they are safe and environment friendly. Although much researches have been done on neem in different aspect such as on uses (Hegde, 1991; Harsh *et al.*, 1992; Gill and Deb Roy, 1993). The neem tree, *Azadirachta indica* A. Juss (Meliaceae), has been found to be a promising source of natural pesticides, several constituents of its leaves and seed showing marked insect control potential. Neem seed

kernel extract suppresses the feeding, growth and reproduction of insects (Schmutterer, 1990; Butterworth and Morgan, 1971).

Keeping the above facts in view, an attempt was made to carry out studies on the effect of neem extract on leafhopper (*A. b.biguttula*) of potato crop (*Solanum tuberosum* L.) in agro-ecological condition of Quetta Baluchistan.

OBJECTIVES:

1. To identify the leafhopper (*Amrasca biguttula biguttula*) species damaging potato crop (*Solanum tuberosum*L.) at Quetta, Baluchistan
2. To evaluate the neem extract on the population of leafhopper (*Amrasca biguttula biguttula*) of potato (*Solanum tuberosum*L.).

CHAPTER-II

REVIEW OF LITERATURE

Johnson *et al.* (1986) presented model parameterization and output plotted against field data for two cultivars, each planted a total of three times in two seasons. The model is intended for use in analyzing the dynamics of yield under variable pest pressure and environmental conditions. Model sensitivity, behavior and use in relation to modeling yield loss due to pests are discussed.

Isman *et al.* (1990) analyzed twelve samples of neem oil by liquid chromatography, azadirachtin concentrations ranged from 50 (limit of detection) to over 4000ppm. Oils were bio-assayed for larval growth inhibition and anti-feedant activity against the variegated cutworm and for molt disrupting activity against the milk weed bug. For each of the three bio-assays, bioactivity of the oils, as measured by EC60 value, is highly correlated with azadirachtin content of the oils.

Isman (2000) reviewed the (i) the range of biological activities of essential oils and their constituents, (ii) their toxicity and proposed model of action in insects, (iii) their potential health and environmental impacts as crop protectants and (iv) commercialization of pesticides based on plant essential oils.

Kennedy and Storer (2000) described that the abilities of polyphagous pest species to move among and utilize different habitat patches in response to changes in suitability enable the pests to exploit unstable cropping systems. These pest characteristics determine the location and timing of damaging population. Habitat suitability is influenced by plant species and cultivar, crop phenology and agricultural inputs. Pest movement is affected by a suit of intrinsic factors, such as population age structure and mobility, and extrinsic factors including weather systems and habitat distributions.

Alam *et al.* (2010) investigated the population abundance and determined the efficacy of three botanical oils i.e. Neem, Mahogani and Karanja and one synthetic insecticide (Admire 200 SL) against okra leafhopper. In evaluating the effectiveness of the botanical oils and insecticide, three times applications were made at seven days interval. The leafhopper population varied significantly with the application of botanicals and insecticide. Admire 200 SL gave the best result among the treatment. Out of three botanicals the Karanja repelled (93.33%), whereas Mahogoni and neem repelled (86.66%) and (63.33%).

Iqbal *et al.* (2011) assessed the effect of six different botanicals i.e. Neem extract (Azadirachtin) Orange peel (*C. sinensis*), Bitter guard (*M. dioica*), Garlic (*A. vineale*), Mari goldl, Hot pepper (*C. frutescens*) and Tobaccso (*N. tabacum*) extract on wheat aphid in field at Adaptive Research Farm Gujranwala. Wheat aphids were deliberately exposed to the above botanical extracts and then the number of live and dead aphids was counted in meter square ring on tagged spikelet.

Akbar *et al.* (2012) considering the importance of organic farming and the demand for safe food globally, evaluated the performance of bio-pesticides (neem formulation & bacterial derived Spinosad) in comparison with conventional insecticides (imidaclopride, endosulfan & profenofos) against potato leafhopper on autumn potato crop (*S. tuberosum*). All the conventional insecticides were found effective against potato leafhopper. Biosal showed moderate effectiveness (55.91%), while Spinosad exerted least effect (15.63%) reduction in potato leafhopper population.

Karim (2012) conducted an experiment at Sher-e-Bengal Agricultural University Dhaka. He evaluated some plant extracts against sucking insect pests such as, aphid, whitefly and mealy bug attacking brinjal. The population of the aforementioned pests decreased with plant age. The lowest number of pests were observed in neem seed karnal extract treated plot. The number of leaves and Branches per plant, number of fruits per plant, length, diameter and weight of individual fruit were found highest in neem seed karnal extract at 20g/l treatment. The same treatment gave the highest yield (29.58t/hac) and increased (41.10%) yield over control. The maximum net return (330071 tk/hac) and the highest benefit cost ratio (2.28) was also observed in neem seed karnal extract.

Adnan *et al.* (2014) examined the efficacy of bio-pesticide against Mango-hopper. Result indicated that Azadirachtin based neem oil found effective against mango hopper as (48.35%), (60.15%) and (56.54%) reduction after 24hrs, 72hrs and 168hrs of the

spray respectively, which was comparable with Cypermethrin. Natural enemies were also higher after 1st and 2nd spray in case of neem oil.

Yousufzai *et al.* (2015) examined that mortalities among the newly hatched nymph influenced significantly ($P < 0.05$) by the use of neem extract. Neem leaf extract had the highest mortality i.e. 17.25 per leaf (86.25%) followed by dhatura leaf extract, Eucalyptus leaf extract and neem solutions were 16.50 (82.50%), 15.75 (78.75%) and 14.00 (70%) respectively, as compared to the lower mortality in control/untreated plot 12.25 (61.25%). The adult's activities of leafhopper/jassid (*A. devastans*) showed that neem leaf extract treated adult's females had lowest fecundity, while fecundity of females prolonged to 9, 11, 14 and 18 eggs when treated with dhatura leaf extract, Eucalyptus leaf extract, neem solution and control/untreated respectively. It is recommended that neem leaf extract and Dhatura leaf extract may preferably be used as to check and control the nymphal development of leafhopper/jassid (*A. devastans*) and to reduce the survival of leafhopper adult on sunflower.

CHAPTER-III

MATERIALS AND METHODS

An experiment was carried out to determine the efficacy of bio-pesticide (Neem extract) against leafhopper/jassid (*Amrasca biguttula biguttula*) on potato crop in the experiment area of Quetta Baluchistan, and it is divided into two parts: laboratory experiments and field trials during the year 2015.

3.1 Laboratory experiments

The trial was carried out at Entomology Department Lab. All experiments were done with the objective of identification and testing the efficacy of neembased products against leafhoppers (*Amrasca biguttula biguttula*) in the field.

3.2 Identification:

Methods of identification: Specimens were identified using identification keys printed on the Lab Manual as well publications.

Method of imaging: Images were captured using high pixel camera. Microscopic examination was done with the help of USB camera (350 k pixel) **a)** Labomed CSM2 (20X and 40X), **b)** Kyowa Medilux 20 will be used.

LAYOUT PLAN OF THE EXPERIMENT

Title: Efficacy of bio-pesticides against leafhopper/jassid (*Amrasca biguttula biguttula*) on potato (*Solanum tuberosum* L.).

Layout: Randomized Complete Block Design.

Replication: Four.

Treatments: Four.

T1 = Neem extract (25cc)

T2 = Neem extract (50cc)

T3 = Neem extract (75cc)

T4 = Neem extract (100cc)

T5 = Control (untreated)

Plot Size: 5X4 meter

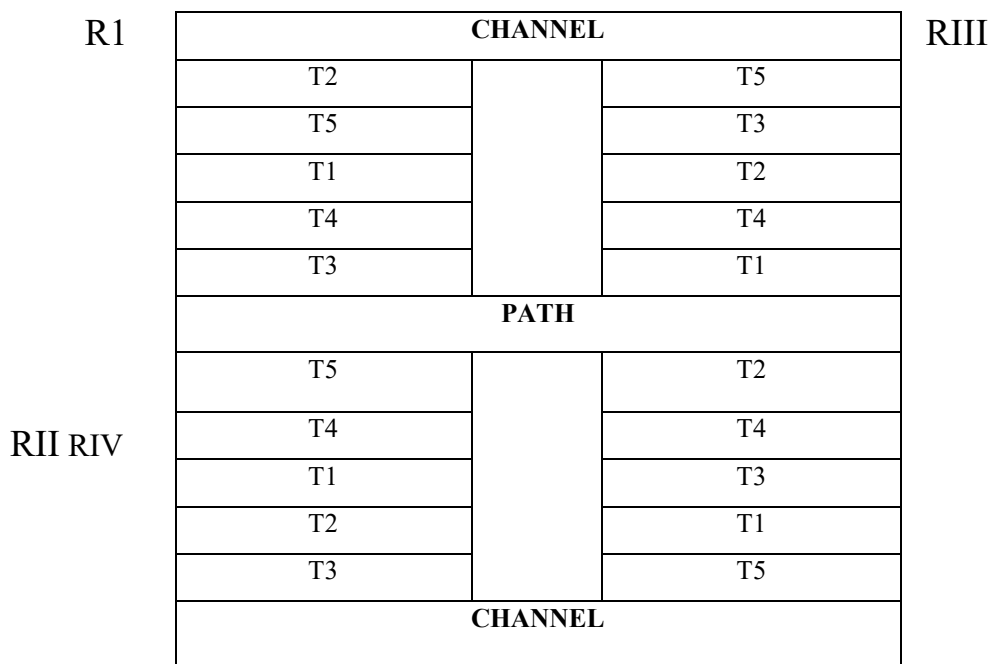


Fig-1. Layout plan of the experiment.

3.3 Evaluation of Neem extract:

Evaluation of Neem extract was carried out at the Directorate of Agriculture Research Farm, during the period of 2015, so that the yield estimate and recommendations for pest control could be applied in the research fields of Research Farm. Healthy buds of white skin potato (varieties: Diamond, krin, Altamash and Curado) crop were buried to the well prepared ridges keeping 75cm distance between ridge/row to ridge/row and 60cm distance between plant to plant during the month of March-April.

Table-1: Bio-pesticides and their doses used against leafhopper/jassid (*Amrasca biguttula biguttula*) on potato crop (*Solanum tuberosum*L.).

Treatment	Bio pesticides used	Dose/ acre	Dose/ plot
T1	Neem extract	1litter/acre	25cc
T2	Neem extract	2liter/acre	50cc
T3	Neem extract	3liter/acre	75cc
T4	Neem extract	4liter/acre	100cc
T5	Control/untreated	0litter/acre	00cc

The above bio pesticides were used according to the ETL level. Spray tank was carefully washed with admixture before pouring bio-pesticides. Overall, 03 sprays were applied at the interval of 1 month in potato field.

Tagging was applied in five plants for each treatment for recording leafhopper population. 05 leaves were observed from top, middle and bottom for insect pest infestation. The pre-treatment observation were noted before 01 day of each spray.

Post treatment observations were noted after interval of 24, 48 and 72 of sprays (Singh and Kushik, 1990).

To see the efficacy of bio-pesticides reduction percentage was calculated according to Henderson-Tilton's (1955) as under;

$$\text{Percentage Mortality} = 1 - \frac{\frac{T_a}{C_a}}{\frac{T_b}{C_b}} \times 100$$

Where;

- Tb = Number of leafhopper/jassid in the treated plots before treatment.
- Ta = Number of leafhopper/jassid in the treated plots after treatment.
- Cb = leafhopper/Jassid population in the control plots before treatment.
- Ca = leafhopper/Jassid population in the control plots after treatment.

3.5 Statistical analysis and evaluation of the results:

Collected data were interpreted in spreadsheet of Microsoft excel. Then further data analysis ANOVA was carried out through software SXW 8.0.

CHAPTER-IV

RESULTS

Identification

For the identification of potato leafhopper species, they were collected and studied at postgraduate systematic laboratory, Sindh Agriculture University Tandojam. Habitus images were captured through microscope, further for the study of genitalia, maceration was undertaken to remove soft bodies and connective tissues to have clear genitalia for study. The study revealed it *Amrasca biguttula biguttula* (Ishida, 1912) the detailed hierarchy is given below and habitus images and genitalia are provided in Plate-1.

Taxonomic Hierarchy of potato leafhopper *Amrasca biguttula biguttula* (Ishida, 1912)

Order: Hemiptera, Linnaeus 1758

Suborder: Auchenorrhyncha, Duméril 1806

Family: Cicadellidae, Latreille 1802

Subfamily: Typhlocybinae, Kirschbaum 1868

Genus: *Amrasca* Ghauri, 1967

Species: *biguttula*

Subspecies: *biguttula*

Author: Ishida

Year: 1912

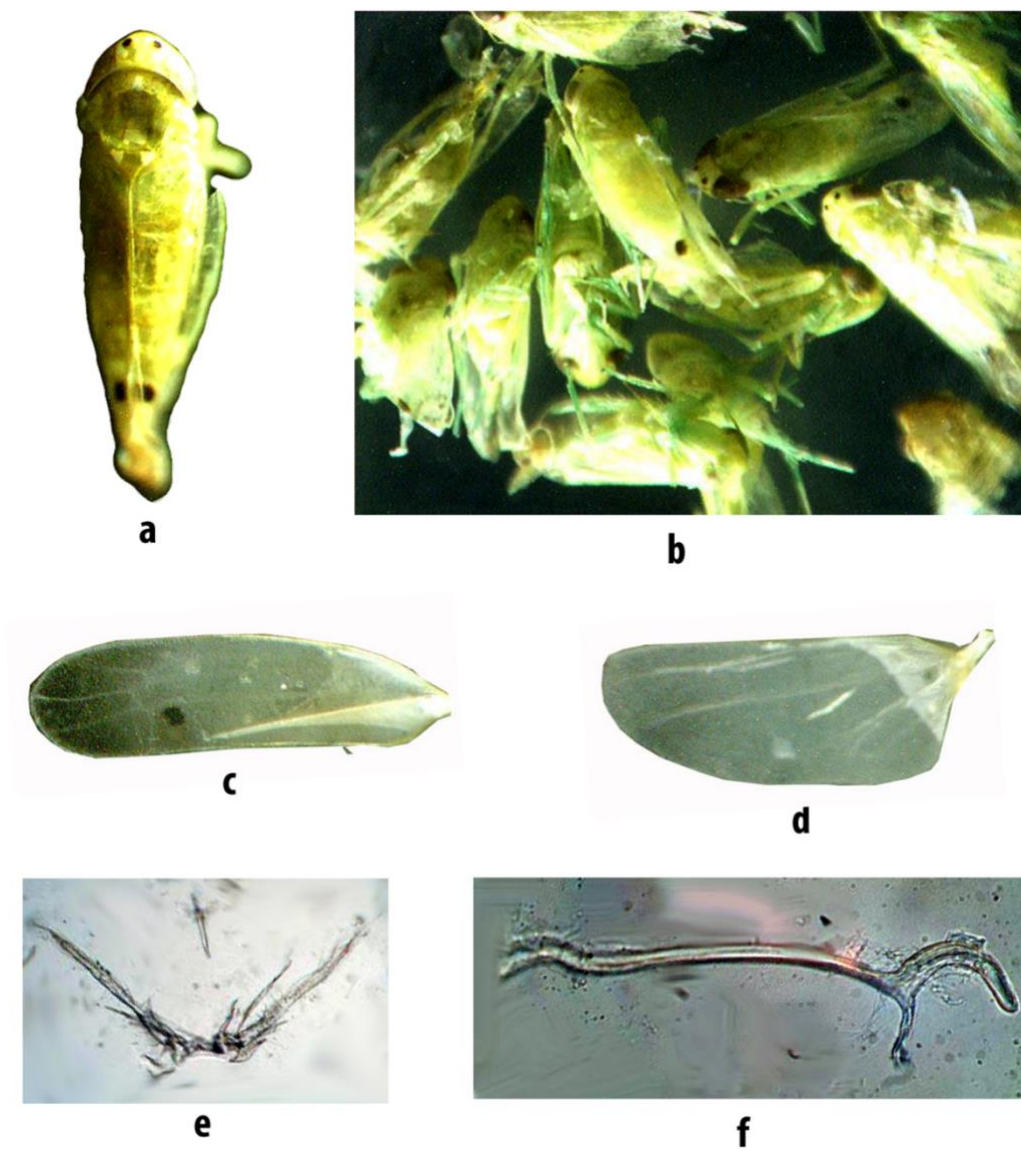


Plate -1. *Amrasca biguttula biguttula* (Ishida, 1912): a, habitus, b, group of collection, c, forewing, d, hindwing, e, male valve, style, aedeagus, f, aedeagus.

Evaluation of neem extract on the population of leafhoppers of potato:

In order to test the efficacy of different dose of neem extract against leafhopper/jassid(*Amrasca biguttula biguttula*) in potato crop (*S.tuberosum*L.), the studies were carried out during the period of 2015. The experimental crop was sown in the experimental area of Quetta, Balochistan. The potato crop (*S. tuberosum*L.) was sprayed by different neem extract doses 1litter/acre, 2 liters/acre, 3 liters/acre and 4 liters/acre for control of leafhopper/jassid(*A. b.biguttula*) on potato crop (*S. tuberosum*L.) and the leafhopper/jassid(*A. b.biguttula*) infestation and neem extract efficacy was compared by control (untreated). Three sprays were carried out when the increase in the leafhopper/jassid(*A. b.biguttula*) infestation was noted and the results are presented in Tables 1 to 3.

1st spray:

Before first spray of different dose of neem extract on potato crop(*Solanum tuberosum*L.), the pre-treatment count of leafhopper/jassid(*A.b.biguttula*) on five leaves of per plant was managed and the data Table-1 showed that the $T_4 = 100\text{cc/plot}$ was most effective to reduced leafhopper/jassid(*A. b.biguttula*)population with highest efficacy (54.53%); where the pre-treatment leafhopper/jassid(*A.b.biguttula*)population (14.98/ 5 leaves) reduced to 8.33, 7.43 and 6.18/5 leaves after 24hrs, 48hrs and 72hrs of first spray, respectively. $T_3=75\text{cc/plot}$ ranked 2nd by efficacy (46.91%) reducing pre-treatment leafhopper/jassid(*A. b.biguttula*)population (15.03/ 5 leaves) to 8.5, 8.18 and 7.25/ 5 leaves after 24hrs, 48hrs and 72hrs of first spray, respectively. The neem extract dose of 50cc/plot ranked 3rd by efficacy (41.13%) reducing pre-treatment leafhopper/jassid(*A. b.biguttula*)population (15.03/ 5 leaves) to 9.33, 9.15 and 8.03/ 5 leaves after 24hrs, 48hrs

and 72hrs of first spray respectively, and dose 25cc/plot of neem extract ranked 4th by efficacy (33.76%) reducing pre-treatment leafhopper/jassid(*A. b.biguttula*) population (15.13/ 5 leaves) to 10.23, 9.6 and 9.1/ 5 leaves after 24hrs, 48hrs and 72hrs of first spray respectively.

Table 2: Mean population/five leaves and reduction %age of leafhopper/jassid (*Amrasca biguttula biguttula*) recorded after different time intervals of 1st

spray with various amounts of Neem extract on potato crop (*Solanum tuberosum*L.).

Treatment/Neem extract/doses/plot	Pre-Treatment	Post treatment observation			Pest Reduction	Reduction n%
		24 hrs	48 hrs	72 hrs		
T1= 25cc/plot	15.13	10.23	9.6	9.1	6.03d	33.76d
T2= 50cc/plot	15.03	9.33	9.15	8.03	7.00c	41.13c
T3=75cc/plot	15.03	8.5	8.18	7.25	7.78b	46.91b
T4=100cc/plot	14.98	8.33	7.43	6.18	8.80a	54.53a
T5= Control/untreated	15.9	14.8	14.23	14.43	1.48e	9.11e

SE±	0.2847	1.5828
LSD 0.05	0.6203	3.4485
LSD 0.01	0.8696	4.8346
CV	6.48	6.04

2nd Spray:

Before second spray of different dose of neem extract on potato crop(*Solanum tuberosum*L.), the pre- treatment count of leafhopper/jassid(*A. b.biguttula*) on five leaves of per plant was managed and the data Table-2 showed that the $T_4 = 100\text{cc/plot}$ was most effective to reduced leafhopper/jassid(*A. b.biguttula*) population with highest efficacy (61.02%); where the pre-treatment leafhopper/jassid(*A. b.biguttula*) population (11.4/ 5 leaves) reduced to 7.925, 5.8 and 4.15/5 leaves after 24hrs, 48hrs and 72hrs of second spray, respectively. $T_3=75\text{cc/plot}$ ranked 2nd by efficacy (52.16%) reducing pre-treatment leafhopper/jassid(*A. b.biguttula*) population (11.675/ 5 leaves) to 8.5, 6.55 and 5.225/ 5 leaves after 24hrs, 48hrs and 72hrs of second spray, respectively. The neem extract dose of 50cc/plot ranked 3rd by efficacy (43.51%) reducing pre-treatment leafhopper/jassid(*A. b.biguttula*) population (11.45/ 5 leaves) to 8.8, 7.25 and 6.075/ 5 leaves after 24hrs, 48hrs and 72hrs of second spray, respectively, and dose 25cc/plot of neem extract ranked 4th by efficacy (34.50%) reducing pre-treatment leafhopper/jassid(*A. b.biguttula*) population (11.53/ 5 leaves) to 9.5, 8.25 and 7.08/ 5 leaves after 24hrs, 48hrs and 72hrs of second spray, respectively.

Table 3: Mean population/five leaves and reduction %age of leafhopper/jassid(*Amrasca biguttula biguttula*) recorded after different

time intervals of 2nd spray with various amounts of Neem extract on potato crop (*Solanum tuberosum* L.).

Treatment/Neem extract/doses/plot	Pre-Treatment	Post treatment observation			Pest Reduction	Reduction%
		24 hrs	48 hrs	72 hrs		
T1= 25cc/plot	11.525	9.5	8.25	7.075	4.45c	34.508d
T2= 50cc/plot	11.45	8.8	7.25	6.075	5.38b	43.51c
T3=75cc/plot	11.675	8.5	6.55	5.225	6.45a	52.16b
T4=100cc/plot	11.4	7.925	5.8	4.15	7.25a	61.02a
T5= Control/untreated	12.3	11.6325	11.2075	11.525	0.78d	6.41e

SE±	0.4023	1.6498
LSD 0.05	0.8766	3.5945
LSD 0.01	1.229	5.0392
CV	11.71	5.9

3rd Spray:

Before third spray of different dose of neem extract on potato crop(*Solanum tuberosum*L.), the pre- treatment count of leafhopper/jassid(*A. b.biguttula*)on five leaves of per plant was managed and the data Table-2 showed that the $T_4 = 100\text{cc/plot}$ was most effective to reduced leafhopper/jassid(*A. b.biguttula*)population with highest efficacy (62.37%); where the pre-treatment leafhopper/jassid(*A. b. biguttula*)population (10.45/ 5 leaves) reduced to 7.115, 4.9 and 3.7/5 leaves after 24hrs, 48hrs and 72hrs of third spray respectively. $T_3=75\text{cc/plot}$ ranked 2nd by efficacy (53.05%) reducing pre-treatment leafhopper/jassid(*A. b.biguttula*)population (10.675/ 5 leaves) to 7.635, 6.0325 and 4.7275/ 5 leaves after 24hrs, 48hrs and 72hrs of third spray respectively. The neem extract dose of 50cc/plot ranked 3rd by efficacy (45.05%) reducing pre-treatment jassid population (10.525/ 5 leaves) to 8.0575, 6.8325 and 5.45/ 5 leaves after 24hrs, 48hrs and 72hrs of third spray respectively, and dose 25cc/plot of neem extract ranked 4th by efficacy (34.42%) reducing pre-treatment leafhopper/jassid(*A. b.biguttula*)population (9.9875/ 5 leaves) to 8.7275, 7.6 and 6.175/ 5 leaves after 24hrs, 48hrs and 72hrs of third spray respectively.

Table 4: Mean population/five leaves and reduction %age of leafhopper/jassid(*Amrasca biguttula biguttula*) recorded after different time intervals of 3rd spray with various amounts of Neem extract on potato crop (*Solanum tuberosum*L.).

Treatment/Neem extract/doses/plot	Pre-Treatment	Post treatment observation			Pest Reduction	Reduction%
		24 hrs	48 hrs	72 hrs		
T1= 25cc/plot	9.9875	8.7275	7.6	6.175	3.81d	34.42d
T2= 50cc/plot	10.525	8.0575	6.8325	5.45	5.08c	45.05c
T3=75cc/plot	10.675	7.635	6.0325	4.7275	5.95b	53.05b
T4=100cc/plot	10.45	7.115	4.9	3.7	6.75a	62.37a
T5= Control/untreated	11.125	10.9775	10.525	10.475	0.65e	5.75e

SE±	0.2124	1.752
LSD 0.05	0.4629	3.8174
LSD 0.01	0.6489	5.3517
CV	6.76	6.17

Mean reduction %age of leafhopper/jassid(*Amrasca biguttula biguttula*)recorded three sprays with various amounts of bio-pesticide on potato crop(*Solanum tuberosum*L.).

After calculating the mean reduction percentage of potato leafhopper(*Amrasca biguttula biguttula*) after the application of all the respective sprays i.e. 1st, 2nd and 3rd sprays, the results revealed that all the sprays of neem extract against potato leafhopper (*Amrasca biguttula biguttula*) remained effective after the observation of leafhopper (*Amrasca biguttula biguttula*) population on 24hrs, 48hrs and 72hrs after the respective spray. T₄=100cc/plot showed highest efficacy (59.31a %) trough out the three sprays followed by T₃=75cc/plot (50.71b %), T₂=50cc/plot (43.23c %) and T₁=25cc/plot (43.23d %) respectively, with comparison to T₀=Control/untreated (7.09e %).

Table 5: Mean reduction %age of leafhopper/jassid(*Amrasca biguttula biguttula*) recorded three sprays with various amounts of bio-pesticide on potato crop (*Solanum tuberosum*L.).

Treatment/Neem extract/doses/plot	1st Spray	2nd Spray	3rd Spray	Mean
T1= 25cc/plot	33.76	34.508	34.42	34.23d
T2= 50cc/plot	41.13	43.51	45.05	43.23c
T3= 75cc/plot	46.91	52.16	53.05	50.71b
T4= 100cc/plot	54.53	61.02	62.37	59.31a
T5= Control/untreated	9.11	6.41	5.75	7.09e

SE±	1.9524
LSD 0.05	4.5022
LSD 0.01	6.551
CV	6.14

CHAPTER-V

DISCUSSION

The results of present study on efficacy of bio-pesticides/neem leaves extract against leafhopper/jassid (*Amrasca biguttula biguttula*) on potato crop (*Solanum tuberosum* L.). Indicated that all four levels of bio-pesticides namely Neem leaves extract applied reduced the population of leafhopper/Jassid (*A. b. biguttula*) after 24hrs, 48hrs and 72hrs of their application during three sprays. However, of Neem leaves extract at 100cc level found superior and caused significantly greater reduction percentage (54.53a%) of the Leafhopper/Jassid (*A. b. biguttula*) population after 1st spray, closely followed by 75cc, while 50cc and 25cc when applied did not prove effective as 100cc.

The results revealed that all the sprays of neem extract against potato leafhopper (*A. b. biguttula*) remained effective after the observation of leafhopper (*A. b. biguttula*) population on 24hrs, 48hrs and 72hrs after the respective spray. T₄=100cc/plot, showed highest efficacy (59.31a %) through out the three sprays followed by T₃=75cc/plot (50.71b %), T₂=50cc/plot (43.23c %) and T₁=25cc/plot (43.23d %) respectively, with comparison to T₅=Control/untreated (7.09e %).

These results are supported by the finding of Akbar *et al.*, (2012) considering the importance of organic farming and the demand for safe food globally, evaluated the performance of bio-pesticides (neem formulation & bacterial derived Spinosad) in comparison with conventional insecticides (imidaclopride, endosulfan & profenofos) against potato leafhopper on autumn potato crop (*S. tuberosum*). All the conventional insecticides were found effective against potato leafhopper. Biosal showed

moderate effectiveness (55.91%), while Spinosad exerted least effect (15.63%) reduction in potato leafhopper population. Karim (2012) evaluated some plant extracts against sucking insect pests such as, aphid, whitefly and mealy bug attacking brinjal. The population of the aforementioned pests decreased with plant age.

The lowest number of pests were observed in neem seed karnal extract treated plot. The number of leaves and Branches per plant, number of fruits per plant, length, diameter and weight of individual fruit were found highest in neem seed karnal extract at 20g/l treatment. The same treatment gave the highest yield (29.58t/hac) and increased (41.10%) yield over control. The maximum net return (330071 tk/hac) and the highest benefit cost ratio (2.28) was also observed in neem seed karnal extract. Adnan *et al.*, (2014) indicated that Azadirachtin based neem oil found effective against mango hopper as (48.35%), (60.15%) and (56.54%) reduction after 24hrs, 72hrs and 168hrs of the spray respectively, which was comparable with Cypermethrin. Natural enemies were also higher after 1st and 2nd spray in case of neem oil. Yousufzai *et al.*, (2015) examined that mortalities among the newly hatched nymph influenced significantly ($P < 0.05$) by the use of neem extract. Neem leaf extract had the highest mortality i.e. 17.25 per leaf (86.25%) followed by Dhatura leaf extract, Eucalyptus leaf extract and neem solutions were 16.50 (82.50%), 15.75 (78.75%) and 14.00 (70%) respectively, as compared to the lower mortality in control/untreated plot 12.25 (61.25%). The adult's activities of leafhopper/jassid (*A. devastans*) showed that neem leaf extract treated adult's females had lowest fecundity, while fecundity of females prolonged to 9, 11, 14 and 18 eggs when treated with Dhatura leaf extract, Eucalyptus leaf extract, neem solution and

control/untreated respectively. It is recommended that neem leaf extract and Dhatura leaf extract may preferably be used as to check and control the nymphal development of leafhopper/jassid(*A. devastans*) and to reduce the survival of leafhopper adult on sunflower.

CHAPTER-VI

SUMMARY, CONCLUSIONS AND RECOMMENDATION

Summary:

A field experiment was carried out to evaluate the efficacy of neem-extract (Azadirachtin) against leafhopper/jassid (*A. b. biguttula*) on potato (*Solanum tuberosum* L.) in the experimental area of Baluchistan Agricultural Research Center Quetta, Baluchistan, during the growing season of 2015.

Potato seed was sown on ridges keeping a ridge to ridge distance of 75 cm. After completion of germination seedlings and a plant to plant distance of 45 cm was maintained. There were twenty plots each plot measuring an area of 5x4 meter. Following neem-extracts were applied thrice against leafhopper/jassid (*A. b. biguttula*) at economic threshold level. The experiment was laid out in a four replicated Randomized Complete Block Design.

Pre-treatment population of leafhopper/jassid (*A. b. biguttula*) was recorded one day before spray from randomized selected 25 plants from each treatment (5 levels each one from top, two from middle and two from bottom). While post treatment leafhopper/jassid (*A. b. biguttula*) count was made after 24hrs, 48hrs and 72hrs of spray.

To see the efficacy of bio-pesticides reduction %age was calculated as per standard method. The data thus collected were subjected to analysis of variance to test the superiority of treatment mean LSD test was applied. The findings of the experiment are given here under;

It was observed that during 1st spray all the four level (neem extracts) bio-pesticides reduced the population of potato leafhopper/jassid(*A. b. biguttula*) as compared to Untreated/Control. However, the reduction percentage varied significantly between the products after 24hrs, 48hrs and 72hrs of their application. Among the bio-pesticides evaluated Neem leaves extract at 100cc proved superior and caused greater reduction of pest after each interval i.e. 24hrs, 48hrs and 72hrs respectively.

It was further found that during that 2nd spray of respective bio-pesticides, the trained of efficacy remained more or less same. Neem extract 100cc remained its effect superior after 24hrs, 48hrs and 72hrs of their application.

During third spray the efficacy of bio-pesticides remained same as recorded in the 1st and 2nd spray respectively. All products showed effective reduction of leafhopper/jassid(*A. b. biguttula*) up to 72hrs of their application then declined. Among the products combination of Neem extract at 100cc recorded maximum reduction of leafhopper/jassid(*A. b. biguttula*) after 24hrs, 48hrs and 72hrs respectively.

On the basis of overall average performance of four products during three spray, it was found that Neem leaves extract at 100cc found effective in controlling the leafhopper/jassid(*A. b. biguttula*) on potato crop, it reduces pest by (59.31a %), followed by 75cc (50.71b %), while other two products i.e., 50cc (43.23c %) and at 25cc (34.23d %) neem leaves extract was less effective against jassid(*A. b. biguttula*) on potato crop (*Solanum tuberosum* L.) with comparison to 00cc/control (7.09e %).

Conclusions

On the basis of the results of present experiment it may be concluded that all four bio-pesticides reduces the population of leafhopper/jassid (*A. b. biguttula*) on potato crop (*Solanum tuberosum*L.) during each spray. However, neem leaves extract at 100cc was found superior in reducing the pest closely followed by neem extract at 75cc, while neem extract 50cc and 25cc level were proved less effective in decreasing the population of leafhopper/jassid(*A. b. biguttula*)on potato crop(*S. tuberosum*L.).

Recommendations

The following recommendations are made:

1. Application of neem leaves extract (Azadirachtin) at 100cc level may be applied at economic threshold lever (ETC) to minimize the infestation of leafhopper/jassid(*A. b. biguttula*)on potato crop(*S. tuberosum*L.).
2. Proper scouting of pest may be made to know the population level of pest.
3. Spray may be done keeping in view the whole plant may be washed out to get fruitful results.
4. Indiscriminate use of bio-pesticides/neem extracts may be avoided; it should be applied on proper time and method to minimize the pest infestation

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APPENDICES

- Appendix-I Analysis of variance for Pest Reductionof leafhopper(*A. b. biguttula*) recorded after of 1st spray with various amounts of Neem extract on potato crop (*Solanum tuberosum*L.)**

Source of variance	DF	SS	MS	F-value	Remarks
Replication	3	7.957	2.6525		
Treatment	4	128.943	32.2357	198.88	0.0000
Error	12	1.945	0.1621		
Total	19	138.845			

Appendix-II Analysis of variance for Reduction% of leafhopper (*A. b. biguttula*) recorded after of 1st spray with various amounts of Neem extract on potato crop (*Solanum tuberosum*L.)

Source of variance	DF	SS	MS	F-value	Remarks
Replication	3	15.60	5.20		
Treatment	4	4843.88	1210.97	241.70	0.0000
Error	12	60.12	5.01		
Total	19	4919.61			

Appendix-III Analysis of variance for Pest Reductionof leafhopper (*A. b. biguttula*) recorded after of 2nd spray with various amounts of Neem extract on potato crop (*Solanum tuberosum*L.)

Source of	DF	SS	MS	F-value	Remarks
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variance					
Replication	3	1.14	0.38		
Treatment	4	101.443	25.3607	78.33	0.0000
Error	12	3.885	0.3238		
Total	19	106.468			

Appendix-IV Analysis of variance for Reduction% of leafhopper (*A. b. biguttula*) recorded after of 2nd spray with various amounts of Neem extract on potato crop (*Solanum tuberosum*L.)

Source of variance	DF	SS	MS	F-value	Remarks
Replication	3	25.06	8.35		
Treatment	4	7037.62	1759.4	323.22	0.0000
Error	12	65.32	5.44		
Total	19	7128			

Appendix-V Analysis of variance for Pest Reductionof leafhopper (*A. b. biguttula*) recorded after of 3rd spray with various amounts of Neem extract on potato crop (*Solanum tuberosum*L.)

Source of variance	DF	SS	MS	F-value	Remarks
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Replication	3	2.4453	0.8151		
Treatment	4	91.078	22.7695	252.27	0.0000
Error	12	1.0831	0.0903		
Total	19	94.6064			

Appendix-VI Analysis of variance for Reduction% of leafhopper (*A. b. biguttula*) recorded after of 3rd spray with various amounts of Neem extract on potato crop (*Solanum tuberosum*L.)

Source of variance	DF	SS	MS	F-value	Remarks
Replication	3	18.59	6.2		
Treatment	4	7602.26	1900.56	309.57	0.0000
Error	12	73.67	6.14		
Total	19	7694.52			

Appendix-VII Analysis of variance for Pest Reduction % age of leafhopper (*A. b. biguttula*) recorded after of all spray with various amounts of Neem extract on potato crop (*Solanum tuberosum*L.)

Source of variance	DF	SS	MS	F-value	Remarks
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Sprays	2	25.89	12.94		
Treatment	4	4824.81	1206.2	210.96	0.0000
Error	8	45.74	5.72		
Total	14	4896.43			

SYNOPSIS

SYNOPSIS OF THE PROPOSED THESIS RESEARCH WORK TO BE DONE IN CONNECTION WITH THE PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF MASTER OF SCIENCE (AGRICULTURE) HONOURS IN ENTOMOLOGY, TO BE SUBMITTED TO SINDH AGRICULTURE UNIVERSITY,

TANDOJAM, BY MR. NASRULLAH S/O MEHRULLAH MASTOI, REG. NO. 2K13-ENT-287

Title: Effect of Neem extract on leafhopper/jassid (*A. b. biguttula*) of potato crop (*Solanum tuberosum* L.) at Quetta, Baluchistan

INTRODUCTION

Potato (*Solanum tuberosum* L.) is the world's leading vegetable crop along with staple food and is placed at fourth position after rice, wheat and maize respectively (Rauscher *et al.*, 2006). Potato has two compensations over the other crops i.e. its production and calories per unit area are higher than wheat and rice while in term of total production potato provides 12-15 times more yield per hectare as compared with cereals. Presently, it is estimated that the total annual domestic production amounts to around 1.9MT (GOP, 2011). In high altitude valley potato is grown as summer crop and in the plains and low valleys three consecutive crops can be grown per year.

Pakistan can be separated into eight agro ecological zones for the production of potato which are: (I) Southern Balochistan and Punjab and Irrigated plains of Sindh; (II) Irrigated plains of South East Khyber Pakhtunkhwa and Central Punjab; (III) Irrigated and rain fed plains of Khyber Pakhtunkhwa and Northern Punjab; (IV) Irrigated lower valleys of Khyber Pakhtunkhwa; (V) Rain fed high valleys and hill sides of Khyber Pakhtunkhwa, Azad Kashmir and Northern Punjab; (VI) Irrigated high valleys of Khyber Pakhtunkhwa, Northern Areas around Azad Kashmir and Chillas; (VII) Irrigated high valleys of Northern Areas and Khyber Pakhtunkhwa around Mastuj; (VIII) Irrigated high valleys of Balochistan and South and North Waziristan. These eight agro ecological zones are different in altitude, longitude, Latitude, topography, climate, soil and irrigation

infrastructure (Zanoni, 1991). Northern Pakistan is the key for seed producing and supplies the seed to the rest of the country.

Nevertheless, the insect pests cause 35-40% losses that may reach up to 60-70% (Salim, 1999). Potato crop is attacked by different insect pests among them the potato leaf hoppers possess considerable importance. About 79 species of leaf hoppers have been found infesting the potato crop (Ramirez and Ramos, 1982). The extent of damage varies depending upon weather conditions, pest population and availability of alternate hosts. Pesticides are quick solution for insect control and are important in increasing solution for insect control and are important in increasing agriculture production (Mehmood *et al.*, 2001). About one third of the pesticides used in Pakistan are applied on fruits and vegetable crops (Hussain *et al.*, 2002). As a crucial component of Integrated Pest Management (IPM) pesticides can increase the yield by reducing the losses caused by insects.

The adult and immature stages of potato leafhopper/jassid (*A. b. biguttula*) are very similar in appearance. They are light green, slender and sedge-shaped in appearance. The adults are approximately 1/8 inch long. Both adults and nymphs have piercing sucking mouthparts on the underside of the head that they insert into the plant to feed on plant juices. The wingless nymphs move quickly backwards and sideways in a crab like fashion on the leaf surface. Eggs are laid in plant tissue and are difficult to observe in the field. Potato leafhopper *Empoasca fabae* (Harris) (Homoptera: Cicadellidae), is a highly polyphagous pest, capable of successful reproduction on 200 plant species in 25 families (Lamp *et al.* 1994).

Extracts from plant origin containing insecticidal properties are indigenously available and are considered comparatively safe for environment & public health. It has been reported that over 2000 plant species belonging to about 170 natural families are known to have insecticidal properties (Feistein, 1952).

In view of the environmental and health issues the biopesticides are holding attention by scientists and researchers as they are safe and environment friendly. Several attempts have been made to manage *Amrascadevastans* (Akbar *et al.*, 2005, 2006, 2007) by biopesticides on different crops other than potato. Neem based pesticides are good source for the control of different insects in the form of neem oil, neem extract and even neem seed water extract (Jacobson, 1988). They are well known for their safety and good performance (Schmutterer, 2002) and have been found effective in controlling pests like jassid (Ambedkaret *al.*, 1999).

Although much researches have been done on neem in different aspect such as on uses (Hegde, 1991; Harsh *et al.*, 1992; Gill and Deb Roy, 1993). The neem tree, *Azadirachta indica* A. Juss (Meliaceae), has been found to be a promising source of natural pesticides, several constituents of its leaves and seed showing marked insect control potential. Neem seed kernel extract suppresses the feeding, growth and reproduction of insects (Schmutterer, 1990; Butterworth and Morgan, 1971)

Keeping the above facts in view, an attempt will be made to carry out studies on the effect of neem extract on potato crop in agro-ecological condition of Quetta Baluchistan.

Objectives

1. To identify the leafhopper (*A. b. biguttula*) species damaging potato crop (*S. tuberosum* L.) at Quetta, Baluchistan
2. To evaluate the neem extract (Azadirachtin) on the population of leafhoppers (*A. b. biguttula*) of potato (*S. tuberosum* L.).

MATERIALS & METHODS

Place of experiment: (1) Quetta, Baluchistan, (2) Insect Systematic Postgraduate Laboratory, Department of Entomology, Faculty of Crop Protection, Sindh Agriculture University Tandojam.

Variety: white skin.

Bio-pesticide: Neem (*Azadirachta indica*).

Identification:

Methods of identification: To identify the specimen to species it is often necessary to examine the male genitalia (Dietrich 2006). The method of preparing genitalia for identification will be followed as described by, knight (1965). The abdomen will be separated and placed separately in 10% Potassium hydroxide overnight at room temperature and heavily sclerotized specimens will be heated for 10 minutes, next day the specimen will be washed in water for five minutes.

Methods of observation: For habitus images the canon camera model Power Shot Sx510 HS will be used, and for the images of genitalia 350 k pixel, USB camera fitted on microscopes **a)** Labomed CSM2 (20X and 40X), **b)** Kyowa Medilux 20 were used.

Evaluation of Neem extract:

The selected varieties will be sown on March 2015. During each sampling date, 25 potato plants from each plot will be randomly selected and the number of leafhopper/jassid (*A. b. biguttula*) per plant will be counted 2 bottom, 2 middle and 1 from top. Neem extracts will be sprayed fortnightly when required. Post treatment data will be recorded after 24, 48, 72

hours of the respective spray on treated and untreated plot. Yield will also be compared on all treatments.

Process to prepare 1 liter of Pesticide from neem:

Material require: 1 liter water, mixing container, 10 ml of black soap, 50 ml neem oil,

Procedure: add soap in 1 liter of water, stir well, mix oil into water soap then mix, transfer mixture to spraying equipment and add water to make up to 15 liters. This bio pesticide will be used within 8 hours

Data analysis: The data will be analyzed through statistical software SXW 8.0, to categorize the effectiveness of selected groups of insecticides.

REVIEW OF LITERATURE

Johnson *et al.* (1986) presented model parameterization and output plotted against field data for two cultivars, each planted a total of three times in two seasons. The model is intended for use in analyzing the dynamics of yield under variable pest pressures and environmental conditions. Model sensitivity, behavior and use in relation to modeling yield loss due to pests are discussed.

Johnson *et al.* (1987) modeled feeding effects of leafhopper on potato in two ways: as a temporary and reversible reduction in the plant's potential net crop growth rate, and as leaf area lost to hopperburn. Model output was also verified against nymphal counts, hopperburn readings, and final yields from an independent field experiment, which resulted in a one-to-one correspondence between predicted and final yields. Computer simulations with nymph infestations at different times of the season indicated that reductions in final tuber yield were differentially effected by leafhopper feeding at different stages of growth.

Isman *et al.* (1990) analyzed 12 samples of neem oil by liquid chromatography, azadirachtin concentrations ranged from <50 (limit of detection) to over 4000 ppm. Oils were bio-assayed for larval growth inhibition and antifeedant activity against the variegated cutworm (*Peridromasauca*) and for molt disrupting activity against the milk-weed bug (*Oncopeltusfasciatus*). For each of the three bioassays, bioactivity of the oils, as measured by EC60 values, is highly correlated with azadirachtin content of the oils.

Kabrick and Backus (1990) studied the feeding behaviors and salivary deposits of the potato leafhopper, *Empoasca fabae* (Harris), on stems of a susceptible cultivar of alfalfa (*Medicago sativa*, L.) in relation to the cellular damage and initiation of hopperburn they cause. An AC electronic monitoring system (EMS) was used to quantify the number and duration of probes made on alfalfa stems. Light and electron microscopy were then used to examine the effects of these probes. Most probes were short in duration as determined by EMS and comprised primarily of multiple-cell laceration behavior (represented by the I_a waveform). Probes containing still-stylet ingestion behavior (represented by the I_c waveform) were longer in duration and occurred less frequently.

Isman (2000) reviewed the (i) the range of biological activities of essential oils and their constituents; (ii) their toxicity and proposed mode-of-action in insects; (iii) their potential health and environmental impacts as crop protectants; and (iv) commercialization of pesticides based on plant essential oils.

Kennedy and Storer (2000) described that the abilities of poly-phagous pest species to move among and utilize different habitat patches in response to changes in suitability enable the pests to exploit unstable cropping systems. These pest characteristics determine the location and timing of damaging populations. Habitat suitability is influenced by plant species and cultivar, crop phenology, and agricultural inputs. Pest movement is affected by a suite of intrinsic factors, such as population age structure and mobility, and extrinsic factors, including weather systems and habitat distribution.

Iqbal *et al.* (2011) assessed the effect of six different botanical extracts- Orange peel (*Citrus sinensis*); Bitter goard (*Momordicadioica*); Garlic (*Allium vineale*); Mari gold; Hot pepper (*Capsicum frutescens*) and Tobacco (*Nicotianatabacum*) extract on wheat aphid in field at Adaptive Research Farm, Gujranwala. Wheat aphids were deliberately exposed to the above botanical extracts and then the number of live and dead aphids was counted in meter square ring on tagged spikelets.

Debashri and Tamal (2012) reviewed the efficacy of *Azadirachtaindica*. Although both synthetic and natural of pesticides are used extensively in the agricultural fields to control crop pests, it is well known that natural pesticides are eco- friendly and are safe to the non-target organisms. The *Azadirachtaindica* A. Juss (neem tree), has long been recognized for its insecticidal properties. Nearly 550 insect pest species are sensitive to azadirachtin, an active compound extracted from the *A. indica* tree. Nowadays pesticides from *A. indica* become very much popular because of their biodegradability, least persistence and least toxic to non-target organisms, economic and easy availability. In India, neem products are effective against various pests of both crop fields as well as stored grains like rice, wheat, corn, legumes, potato, tomato, etc. This review put a light on the use and efficacy of *A. indica* based pesticides against various pests of both crop fields as well as stored grains of India.

Glare *et al.* (2012) carried out work on bio pesticides and stated that these based on living microbes and their bioactive compounds have been researched and promoted as replacements for synthetic pesticides for many years. However, lack of

efficacy, inconsistent field performance and high cost have generally relegated them to niche products. Recently, technological advances and major changes in the external environment have positively altered the outlook for bio pesticides. Significant increases in market penetration have been made, but bio pesticides still only make up a small percentage of pest control products. Progress in the areas of activity spectra, delivery options, persistence of effect and implementation have contributed to the increasing use of bio pesticides, but technologies that are truly transformational and result in significant uptake are still lacking.

Wright *et al.* (2014) described that potato leafhoppers (*A. b. biguttula*) damage plants by their feeding, which causes a condition referred to as Hopper burn. The first symptom is a brown triangular area at the tip of an infested leaflet. With time, the injured area expands backward into the leaflet. It is not known whether potato leafhoppers inject a toxin when they feed or whether the damage is caused by direct physical blockage of the phloem vessels at feeding sites. Both adult and nymphal leafhoppers injure potato leaves when feeding, but late instar nymphs cause twice as much yield loss as an equal number of adults. Potato leafhoppers are not known to transmit any diseases. Adult potato leafhoppers are best monitored using a sweep net. Nymphal leafhoppers are monitored by examining potato leaves. A minimum sample size of 25 sweeps per field for adult leafhoppers or 35 leaves per field for nymphs is recommended.

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